

EGERTON



UNIVERSITY

UNIVERSITY EXAMINATIONS

REGULAR -NJORO CAMPUS

SECOND SEMESTER, 2019/2020 ACADEMIC YEAR

**THIRD YEAR RE-SIT/SPECIAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN ENVIRONMENTAL SCIENCE**

ENSC 306: ENVIRONMENTAL HEALTH

STREAM: BSc. (ENSC & NARE)

TIME: 2 HRS

EXAMINATION SESSION: AUGUST

YEAR: 2021

INSTRUCTIONS:

- (i) Attempt all questions in Section A and any TWO in Section B
- (ii) Do not write on the question paper
- (iii) Illustrate your answers with suitable diagrams and formulas where appropriate

SECTION A (40 MARKS)

Question One

Write short notes on the following concepts as used in Environmental Health;

- a) Personal and ambient environment **(2 Marks)**
- b) Fundamental principle of toxicology **(4 Marks)**
- (a) Teratogenesis **(4 Marks)**

Question Two

- a) Differentiate between antagonistic and synergistic effects of chemicals and give examples in each case. **(4 Marks)**
- b) What is indoor air pollution? Give three sources of indoor air pollutants and their effects **(6 Marks)**

Question Three

- (a) Explain what you understand by the terms acute and chronic toxicity
(3 Marks)
- (b) With the help of a diagram, explain what you understand by $L.D_{50}$ and NOAEL
and show them in the dose-response curve
(7 Marks)

Question Four

- a) Many factors impact sensitivity to a pollutant in human beings. Explain these factors and
give examples in each case. (6 Marks)
- b) Children are more highly exposed to toxicants than adults. Why? (4 Marks)

SECTION B (30 MARKS)

Question Five

- a) Describe factors that affect dermal absorption of a toxic substance. (6 Marks)
- b) Discuss the major routes of elimination of a toxicant in animals. (9 Marks)

Question Six

In addition to exercising care in processing, the prevention of foodborne illnesses requires an effective sanitation programme. Describe in detail the components of an effective sanitation program aimed at preventing foodborne illnesses associated with biological agents and chemical intoxication
(15 Marks)

Question Seven

Dose-response assessment is an important step in risk assessment with respect to environmental toxicants. Describe in detail the various steps to be considered in dose-response assessment.
(15 Marks)
